

THE TWO-DIMENSIONAL CHIRAL HIGGS MODEL

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Received 1 August 1991

We show that the two-dimensional chiral Abelian Higgs model, though anomalous, can be consistently quantized provided that gauge degrees of freedom are adequately taken into account. In this way, the anomaly is canceled by a Wess-Zumino term and as a result, instanton contributions are eliminated from the partition function.

Topological objects like instantons and solitons may have a significant impact on the theory in which they arise. In particular, non-trivial structure of the vacuum state generated by instantons in gauge theories may lead to confinement of a fundamental quantum number, anomalous non-conservation of fermion quantum number and other interesting non-perturbative effects.¹

Recently, it has been pointed out that certain topological effects may be wiped out as a result of the lack of gauge invariance arising in the fermionic sector.² This was, in particular, shown for the case of compact quantum electrodynamics in 2+1 dimensions; in this model, fermions generate a Chern-Simons term (responsible for the breakdown of time reversal invariance) which eliminate instanton (monopole) contributions from the partition function. Indeed, the Chern-Simons term is not gauge-invariant in the presence of monopoles and then integration over the gauge group eliminates all topologically non-trivial sectors from the path-integral.^{2,3}

We investigate in the present work whether the same phenomenon occurs in the two-dimensional Abelian Higgs model coupled to chiral fermions. This is an anomalous model in the sense that the fermion determinant cannot be defined in a gauge-invariant way (see, however, Ref. 4) and hence the associated fermionic current is not conserved. Nevertheless, as it happens with the chiral Schwinger model,^{5,6} it can be consistently quantized provided gauge degrees of freedom are appropriately handled.^{7,8} In addition, the Abelian Higgs model also has instanton solutions - the time honored Nielsen-Olsen vortices⁹ - whose effects have been extensively discussed both in the study of confinement mechanisms¹⁰⁻¹² and fermion number violation.^{13,14} It is then worthwhile to analyze if, as it happens in the three-

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dimensional case, topologically non-trivial sectors are eliminated from the partition function due to the existence of an anomaly. The outcome of our investigation is that this is precisely the case: the anomaly prevents the trivial factorization of the gauge group degrees of freedom; instead, integration over these degrees of freedom leads to a consistent, unitary and Lorentz invariant quantum theory and eliminates instanton contribution from the partition function.

We start by recalling the main arguments leading to consistent quantization of anomalous gauge theories.^{7,8} In the Lagrangian formulation of the path-integral approach to quantization of gauge theories the partition function Z is an integral over the whole space of connections rather than an integral over orbit space. When Dirac fermions are present, there is no harm in looking at Z as an integral over orbit space since a careful definition of the path-integral measure leads to a gauge-group volume factorization. On the contrary, when fermions are Weyl fermions, this factorization does not take place anymore: a proper treatment of gauge degrees of freedom uncovers the presence of a Wess–Zumino term¹⁵ which precisely renders the quantum theory consistent. To see this mechanism at work in the two-dimensional Abelian Higgs model coupled to (left-handed) Weyl fermions, let us start by writing the corresponding Euclidean action:

$$S[A, \phi, \bar{\psi}, \psi] = S_{\text{Higgs}}[A, \phi] + S_{\text{fermions}}[A, \bar{\psi}, \psi], \quad (1)$$

$$S_{\text{Higgs}}[A, \phi] = \int d^2x \left[-\frac{1}{4} F_{\mu\nu}^2 + \frac{1}{2} |(\partial_\mu - ieA_\mu)\phi|^2 + V(\phi) \right], \quad (2)$$

$$S_{\text{fermions}}[A, \bar{\psi}, \psi] = \int d^2x \bar{\psi} D[A] \psi, \quad (3)$$

$$D[A] = (i\rlap{\not{\partial}} + e\rlap{\not{A}}) \frac{1}{2} (1 - \gamma_5). \quad (4)$$

Here A_μ is a U(1) gauge field, ϕ a charged scalar, $V[\phi]$ a symmetry breaking potential (ensuring the existence of vortex-like solutions).

Since $\bar{\psi}, \psi$ are (left-handed) Weyl fermions, the Dirac operator $D[A]$ does not define an eigenvalue problem since it maps negative chirality spinors into positive chirality spinors. This chirality flip problem is usually overcome by defining

$$\det D[A] \equiv \det \hat{D}[A]_{\text{reg}}, \quad (5)$$

where

$$\hat{D}[A] = D[A] + i\rlap{\not{\partial}} \frac{1}{2} (1 + \gamma_5) \quad (6)$$

and $|\text{reg}$ means that an appropriate regularization scheme has been adopted to make sense from the (originally unbounded) product of eigenvalues defining the determinant. Now, the addition of free right-handed fermions solves the chirality-flip problem but creates a new one: under a gauge transformation

$$A_\mu \rightarrow A_\mu^\theta = A_\mu + \frac{1}{e} \partial_\mu \theta, \quad (7)$$

the Dirac operator $\hat{D}[A]$ does not transform as a covariant derivative and then one has in general

$$\det \hat{D}[A^\theta] |_{\text{reg}} \neq \det \hat{D}[A] |_{\text{reg}}. \quad (8)$$

Hence, the fermionic determinant is gauge-dependent and the associated gauge current is anomalous. There are nevertheless several proposals to consistently quantize anomalous theories.⁶⁻⁸ The idea is as follows. Consider the partition function for the model defined by action (1)–(3)

$$Z = \int D\phi DA_\mu D\bar{\psi} D\psi \exp(-S[A, \phi, \bar{\psi}, \psi]). \quad (9)$$

The measure DA_μ on the right-hand side is a measure *over all gauge fields* which eventually reduces, through gauge-group volume factorization, to a measure over orbit space whenever the anomaly is absent. But in the present case we know there is an anomaly and in fact following the standard Faddeev–Popov procedure (see Ref. 16 for a geometrical setting) one ends with

$$Z = \int D\phi DA_\mu \Delta[A] \delta(F[A]) D\theta D\bar{\psi} D\psi \exp(-S[A^\theta, \phi^\theta, \bar{\psi}, \psi]), \quad (10)$$

where $F[A^f] = 0$ is the gauge-fixing condition selecting one representative A^f over each orbit and $\Delta[A^f]$ is the corresponding Faddeev–Popov determinant related to the natural metric over orbit space $\Gamma[A^f]$ and the scale $\rho[A^f]$ of the orbit through the formula¹⁶

$$\Delta[A^f] = \rho[A^f] \sqrt{\det \Gamma[A^f]}. \quad (11)$$

Finally $D\theta$ is the volume element on the group of gauge transformations. Usually, one eliminates the θ -dependence in the integrand of the right-hand side of Eq. (11) by performing the fermionic change of variables

$$\psi \rightarrow \psi' = \exp(-i\theta)\psi, \quad (12)$$

$$\bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} \exp(i\theta). \quad (13)$$

Now, one can easily see that, because of property (8) (responsible for the anomaly) the path-integral measure for Weyl fermions is not invariant under transformations (12) and (13). Indeed, given

$$d\mu_A = \exp(-S_{\text{fermions}}[A, \bar{\psi}, \psi]) D\psi D\bar{\psi}, \quad (14)$$

one has

$$d\mu_{A^\theta} = J[A, \theta] d\mu_A. \quad (15)$$

Here $J[A, \theta]$ is the “Fujikawa Jacobian”¹⁷ which can be defined from (15):

$$J[A, \theta] = \det \hat{D}[A^\theta] / (\det \hat{D}[A]). \quad (16)$$

Then, if one performs the change of variables (13), the partition function becomes

$$Z = \int D\phi DA_\mu \Delta[A] \delta(F[A]) D\theta D\bar{\psi} D\psi J[A, \theta] \exp(-S[\phi, A, \bar{\psi}, \psi]). \quad (17)$$

As we shall see, the presence of the Jacobian J induces a Wess–Zumino¹⁵ action which in fact absorbs the anomaly rendering the quantum theory gauge invariant. Indeed, if one defines the fermionic effective action (which coincides with the fermionic determinant for non-anomalous theories) in the form:

$$\exp(-S_{\text{eff}}[A]) \equiv \int D\theta D\bar{\psi} D\psi J[A, \theta] \exp(-S_{\text{fermions}}[A, \bar{\psi}, \psi]), \quad (18)$$

one can easily verify, using the one-cocycle⁶ condition satisfied by $\log J[A, \theta]$,

$$\log J[A, \theta + \omega] = \log J[A, \theta] + \log J[A^\theta, \omega], \quad (19)$$

that S_{eff} is gauge-invariant:

$$S_{\text{eff}}[A^\omega] = S_{\text{eff}}[A]. \quad (20)$$

This result implies that the fermionic current, defined as $\delta S_{\text{eff}}/\delta A_\mu$ is conserved. We shall then take as the partition function for the chiral Abelian Higgs model:

$$Z = \int D\phi DA_\mu \Delta[A] \delta(F[A]) \exp(-S_{\text{Higgs}}[\phi, A] - S_{\text{eff}}[A]). \quad (21)$$

As stated above, identity (20) guarantees conservation of the fermionic current and this indicates that the theory with partition function Z given by Eq. (21) should be consistently quantized. Of course, in general, unitarity and renormalizability must be investigated. For the two-dimensional chiral Schwinger model and its non-Abelian extension,¹⁸ it has been shown (see Ref. 19 and references therein) that the proposal of Refs. 7 and 8 leads to a consistent, unitary and Lorentz invariant quantum theory both in the path-integral and canonical quantization approaches.²⁰ As we shall see below, the same holds for the chiral Higgs model in two dimensions.

What about instanton contributions to the theory with partition function (21)? As stated above, the Abelian Higgs model has vortex-like solutions⁹ which can be taken as instantons in the computation of non-perturbative effects.^{10–12} Asymptotically, a Nielsen–Olesen vortex configuration takes the form:

$$\lim_{r \rightarrow \infty} A_\mu^{\text{vortex}}(r, \varphi) = \frac{n}{e} \partial_\mu \varphi, \quad (22)$$

$$\lim_{r \rightarrow \infty} \phi^{\text{vortex}}(r, \varphi) = \exp(in\varphi) \phi_0, \quad (23)$$

with $\phi = \phi_0$ the value at which $V[\phi]$ takes its minimum. Such a configuration carries n units of magnetic flux (i.e., it has a topological charge equal to n):

$$\frac{e}{2\pi} \oint A_\mu^{\text{vortex}} dx^\mu = n. \quad (24)$$

The path-integral can then be performed in each topological sector (a configuration satisfying (22) and (23) being a representative in each sector). Using a superscript “ n ” to indicate the topological sector to which Higgs and gauge fields belong, the partition function will then be written in the form:

$$Z = \sum_n \int D\phi^{(n)} DA_\mu^{(n)} \Delta[A^{(n)}] \delta(F[A^{(n)}]) \exp(-S_{\text{Higgs}}[\phi^{(n)}, A^{(n)}] - S_{\text{eff}}[A^{(n)}]). \quad (25)$$

A comment on zero modes of the Dirac operator and the fermionic determinant appearing in the effective action (18) is here in order. We know that the operator $\not{D}[A^{(n)}]$ acting on Dirac fermions has $|n|$ square integrable zero modes. For $n > 0$ ($n < 0$) these zero modes are right-handed (left-handed).²² Hence, the corresponding regularized fermion determinant vanishes for all $n \neq 0$. This automatically ensures that only the $n = 0$ sector contributes to Z for Dirac fermions.²³ On the contrary, the operator $\hat{D}[A^{(n)}]$ has only left-handed zero modes since in the right-handed sector it coincides with the free Dirac operator which does not have normalizable zero modes. Then $\det \hat{D}[A^{(n)}]$ vanishes only in the case $n < 0$. Consequently Z reduces to

$$Z = \sum_{n \geq 0} \int D\phi^{(n)} DA_\mu^{(n)} \Delta[A^{(n)}] \delta(F[A^{(n)}]) \exp(-S_{\text{Higgs}}[\phi^{(n)}, A^{(n)}] - S_{\text{eff}}[A^{(n)}]). \quad (26)$$

In order to pursue our calculation we need now an explicit form for $S_{\text{eff}}[A^{(n)}]$. To this end, we quote the result for the Jacobian J appearing^{7,8} in (18)

$$\log J[A, \theta] = -\frac{1}{4\pi} \int d^2x \left[\frac{(a-1)}{2} \partial_\mu \theta \partial_\mu \theta + e(a-1) A_\mu \partial_\mu \theta + e \varepsilon_{\mu\nu} A_\mu \partial_\nu \theta \right]. \quad (27)$$

Here a is a real parameter introduced by taking into account regularization ambiguities which arise when computing gauge non-invariant determinants.^{5,21} Since we are working in topologically non-trivial sectors, surface terms should not be dropped when performing the θ -integral in S_{eff} . Indeed, gauge transformations θ not vanishing at infinity are compatible with any imposed boundary condition at $r = \infty$ since, for example, $\theta = 2\pi$ is equivalent to $\theta = 0$. This gives, for the second term in the argument of the exponential in (25)

$$e(a-1) \int d^2x A_\mu^{(n)} \partial_\mu \theta = e(a-1) \int d^2x \theta \partial_\mu A_\mu^{(n)} + e(a-1) \oint dx^\mu \theta A_\mu^{(n)} \quad (28)$$

or

$$e(a-1) \int d^2x A_\mu^{(n)} \partial_\mu \theta = e(a-1) \int d^2x \theta \partial_\mu A_\mu^{(n)} + e(a-1) 2\pi n \theta_\infty, \quad (29)$$

where we have used (24) and called θ_∞ the value of θ at infinity. We then see that for $n \neq 0$ surface terms do contribute in a non-trivial way. With all this, we get

after integrating over θ

$$\exp(-S_{\text{eff}}[A^{(n)}]) = \int D\bar{\psi} D\psi \exp[-(S_{\text{fermions}}[A^{(n)}, \bar{\psi}, \psi] + \frac{e^2}{8\pi(a-1)} \int d^2x A_\mu A_\mu + F^{(n)})]. \quad (30)$$

The second term in the argument of the exponential is the usual one obtained (in the Lorentz gauge) after integration over θ when non-trivial topological sectors are not taken into account.^{7,8} The third term precisely corresponds to the border contribution and is given by

$$F^{(n)} = \mathcal{N} \lim_{R \rightarrow \infty} F^{(n)}(R), \quad (31)$$

where

$$F^{(n)}(R) = \frac{n^2 a^2}{8(a-1)} \log R, \quad (32)$$

and $\mathcal{N}$ is a constant.

Then, after taking the limit $R \rightarrow \infty$ we have

$$\exp(-S_{\text{eff}}[A^{(n)}]) = \exp(-S_{\text{eff}}[A^{(0)}]) \delta_{n,0} \quad (33)$$

and hence the partition function Z (Eq. (26)) only picks contribution from the $n = 0$ sector:

$$Z = \int D\phi^{(0)} D A_\mu^{(0)} \Delta[A^{(0)}] \delta(F[A^{(0)}]) \exp(-S_{\text{Higgs}}[\phi^{(0)}, A^{(0)}] - S_{\text{eff}}[A^{(0)}]). \quad (34)$$

As announced in the introduction, the chiral Higgs model, though anomalous can then be consistently quantized and only the $n = 0$ sector contributes to the partition function. The value of $S_{\text{eff}}[A]$ can be easily evaluated from (18). The answer (in the Lorentz gauge) is^{7,8}

$$S_{\text{eff}}[A] = \frac{a^2}{8\pi(a-1)} \int d^2x A_\mu A_\mu. \quad (35)$$

Inserting this value in (34) we see that the result coincides with that corresponding to the Abelian Higgs model with *Dirac* fermions except for the fact that the vector meson mass m_v is given by

$$m_v^2 = e^2 (|\phi_0|^2 + a^2/4\pi(a-1)). \quad (36)$$

Then, Z in (34) defines a unitary, positive model for any value of the parameter a such that $m_v^2 > 0$. The existence of a whole range of the undetermined parameter a for which the model is consistent, Lorentz invariant and unitary with a vector meson mass which is not fixed by gauge invariance (as it happens for models with Dirac

fermions) is typical of anomalous gauge theories at least in $d = 2$ dimensions. One can think that Z defines a family of quantum theories and that the ultimate value for a should be determined by physical considerations (see Ref. 24 for a discussion on these facts).

In summary, we have shown in this work that the chiral Abelian Higgs model in two dimensions, though anomalous, can be consistently quantized following the proposal of Refs. 7 and 8. The anomaly is canceled by a “Wess–Zumino” term, as suggested in Ref. 6 and as a result, instanton contributions are eliminated from the partition function defining the quantum theory. The procedure can be straightforwardly generalized to non-Abelian models and one can also envisage the analysis of four-dimensional Higgs models. However, in this last case the issues of unitarity and renormalizability should be carefully investigated. We hope to report on this problems in the future.

Acknowledgments

We wish to thank D. Cabra, R. E. Gamboa Saraví and J. Solomin for a helpful discussion.

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